

ML Specification Report — playground-series-s3e16

Crab Age Prediction · our from-scratch agent (GBDT pool + tree-search)

Figures grounded in the competition's `facts.json`, `eda_summary.json`, `STATUS.md`, and `experiments_tree_v3.json`.

Overview

The task is to predict the age of crabs from physical measurements (metric: MAE, lower is better). We solve it with a from-scratch pipeline — a pool of gradient-boosted trees (LightGBM / XGBoost / CatBoost) plus diverse variants, refined by a tree search whose weight optimisation is decided directly on the **rounded** integer metric. Our champion is an 8-member blend (tree-search node #15) scoring **rounded OOF MAE 1.33563** on 5-fold CV, submitted to the real leaderboard at Public 1.34315 / Private 1.33859. Against the NVIDIA reproduce-agent (which verbatim-copies a public XGBoost kernel scoring 1.3387), our agent's 1.3356 wins — the lower MAE, so **our agent is the winner** on this episode.

Why it matters. Crab age normally requires destructive manual inspection; predicting it from external measurements supports sustainable fisheries management and harvesting decisions.

Data

Purpose of Data. Predict the integer age of a crab (`Age`) from morphometric measurements — a **regression** Playground Series episode scored by mean absolute error. **Data Format** is clean **tabular CSV**: one categorical column (`Sex` $\in \{I = \text{infant}, M, F\}$) and seven continuous size/weight measures. **Data Volume** is **74,051 training rows / 49,368 test rows**, with 8 raw predictive features once `id` and the target are removed.

Data Quality is high: **no missing values** in either split, **0 duplicate rows**, and near-identical train $\leftrightarrow$ test distributions (all per-feature mean shifts $< 0.4\%$, so no covariate shift). The one raw-data wrinkle is that **Height = 0 in 24 training rows** (physically invalid) — treated as missing and imputed with the median. Severe collinearity is the defining structural feature: `Length` $\leftrightarrow$ `Diameter` $r = 0.989$, and `Weight` $\approx$ `Shucked` + `Viscera` + `Shell` ($r = 0.993$). The target is a right-skewed integer count (**range 1–29, mean 9.97, std 3.18, skew 1.09**, 28 unique values), which is exactly what makes rounding of the regression output matter (see the Inference section). The strongest EDA signals were:

- **Sex='I' (infant)** $\rightarrow$ mean age 7.6 vs M 10.9 / F 11.3 — a strong signal captured by an `is_infant` flag.
- **Shell Weight** is the best-correlated feature with Age (Spearman 0.74); `Shucked Weight` the weakest (0.61).
- **Collinearity** among all size/weight columns (0.86–0.99), motivating a feature-pruned member that drops `Weight`.

- **Integer, right-skewed target** — rounding OOF predictions to the nearest integer improves MAE materially.

Annotation Guidelines. The label is `Age`, a positive integer count. Submissions are real-valued regression predictions scored by MAE; because the ground truth is integer-valued, the post-processed predictions are **rounded to the nearest integer and clipped to ≥ 1** before submission (see Decision Threshold).

Feature Set. From the 8 raw columns we engineer **24 features**, grouped below:

Group	Features
Raw	<code>Length</code> , <code>Diameter</code> , <code>Height</code> , <code>Weight</code> , <code>Shucked Weight</code> , <code>Viscera Weight</code> , <code>Shell Weight</code>
Flags	<code>Height_was_zero</code> , <code>is_infant</code> , <code>is_male</code> , <code>is_female</code>
Weight ratios	<code>Shucked Weight_ratio</code> , <code>Viscera Weight_ratio</code> , <code>Shell Weight_ratio</code> , <code>weight_resid</code> (= <code>Weight</code> - <code>Σparts</code>), <code>parts_sum</code>
Size ratios	<code>diam_len</code> , <code>height_len</code> , <code>height_diam</code>
Derived	<code>volume</code> , <code>density</code> , <code>shell_density</code> , <code>meat_to_shell</code> , <code>shucked_to_shell</code>

Splitting strategy. A single canonical **5-fold StratifiedKFold on binned `Age` (`ages` ≥ 20 merged into one bin)**, **shuffle, seed = 42**, shared across all members so scores are directly comparable. Fold MAE is very stable (1.34–1.37), and the CV scheme is validated against the real leaderboard twice: both real submissions show the same-sign, same-order-of-magnitude CV $\leftrightarrow$ LB gap (rounded OOF ~ 0.003 – 0.0075 better than LB), so **CV is trustworthy**.

Models & Architecture

Purpose of Architecture. A regressor that minimises MAE on the integer crab-age target. **Architecture Type** is a **gradient-boosted decision-tree ensemble** — LightGBM (L1 objective), XGBoost (`reg:absoluteerror`) and CatBoost (MAE) as the core learners, extended with diverse variants (Optuna-tuned LGB, a seed-bagged copy, a boundary-pushed low-LR LGB, a Tweedie-objective LGB, and a feature-pruned LGB that drops the collinear `Weight`), combined by a weighted blend. The champion is an **8-member blend** discovered at tree-search node #15.

Each member takes the same **Input Format** — a numeric feature matrix with `Sex` one-hot encoded — of **Input Dimension 24 features** per row (a 1-D vector; there is no spatial/sequence structure).

Architecture Description. Each GBDT member is an additive ensemble of decision trees grown by histogram-based gradient boosting (representative LGB: `num_leaves` = 63, `n_estimators` up to 3000 with early stopping, `learning_rate` = 0.02). The blend layer sits on top as a convex weighted average of the members' out-of-fold predictions, with weights found by a Dirichlet ($k = 800$) plus coordinate-ascent search scored **directly on the rounded MAE** (`metric_fn = MAE(round(clip(blend)))`). The champion places dominant weight on CatBoost (**0.5061**), followed by the feature-pruned LGB (0.2662) and the boundary-pushed LGB (0.1117). **Model Complexity** is best expressed as trees $\times$ leaves rather than a dense parameter count: each LGB member is up to $\sim 3000 \times 63$ leaves, and the champion blends 8 such members.

Training procedures

Training proceeds from a linear-champion baseline through a tree search on the shared 5-fold split, each stage measured on the same OOF scale (all scores are rounded MAE unless noted):

Stage	Configuration	Rounded OOF MAE
Baseline	LGB/XGB/CAT 0.6/0.3/0.1 blend, coarse simplex grid (exp #1)	1.33812
Phase B	+ Optuna-tuned LGB / + seed-bagged LGB (raw improved, rounded worsened)	1.33850 → 1.33893
Tree search	reproduce 3-way with k=800+coordinate-ascent (node #9)	1.33640
Tree search	+ LGB_tuned / seed2024 / LGBBOUND / TWEEDIE / FEATPRUNE members	1.33635 → 1.33567
Champion	8-way blend, node #15 (weights decided on rounded MAE)	1.33563

The headline training finding is a genuine **raw-vs-rounded inversion**: node #15 has raw OOF MAE 1.35712 (worse than the baseline's 1.35589) yet rounded OOF MAE 1.33563 (better than 1.33812) — the cleanest demonstration that raw and rounded MAE are different optimisation targets here, and every decision must be made on the rounded number.

The **Loss Function** is per-member absolute-error / MAE (LightGBM `regression_l1`, XGBoost `reg:absoluteerror`, CatBoost `MAE`), with model and blend selection on rounded MAE. The **Optimization Algorithm** is gradient boosting (histogram-based) — **not** SGD/ADAM — with blend weights optimised by Dirichlet sampling + coordinate ascent. The **Learning Rate** is the boosting shrinkage (representative LGB `learning_rate` = 0.02; the boundary-pushed member uses 0.005); there is **no Learning Rate Scheduler** (N/A — *GBDT convergence is governed by CV early-stopping, not a schedule*) and **no Batch Size** (N/A — *full-dataset histogram boosting, not mini-batched*).

Training Duration for the tree search was **~1,409 s of wall-clock over 24 evaluated node fits**; individual solo members fit in ~24–37 s each (CatBoost 24 s, LGB 31 s, XGB 37 s). **Training Memory** was **not recorded** — a 74k × 24 float matrix is trivial on the single arm64 CPU machine, peak not separately measured. **Transfer Learning** is N/A (*no pretrained weights*); its analogue here is **cross-competition experience injection** — the distilled "Optuna fold-proxy tune → add to pool → seed-bag" recipe carried in from earlier episodes (which, notably, this episode showed does *not* reliably transfer to a rounded-integer-MAE metric). **Data Augmentation** is N/A (*tabular*); the analogues are feature engineering (the 24 features) and seed-bagging.

Reproducibility Standards are strict: a fixed fold seed (42), logged Optuna/seed-bag seeds, and a rebuild gate that retrains every champion member from scratch and requires digit-for-digit OOF reproduction. The **faithful replay reproduced 24/24 member fits exactly ($\Delta = 0$)**, and the blend reproduces 1.33563 exactly (after recovering the full-precision weight vector — the 4-decimal stored weights alone drift the rounded score to 1.335796).

Inference procedures

Decision Threshold is the **integer round/clip rule**: raw blended predictions are **rounded to the nearest integer and clipped to ≥ 1** (`round(clip(blend))`) before submission — this is the load-bearing post-

processing step, since the target is integer-valued and rounding is what converts the raw-MAE ordering into the far better rounded-MAE score. The champion submission's integer predictions fall in [4, 19] (train range [1, 29]). **Inference Duration** was **not recorded**; GBDT scoring of the 49,368 test rows is sub-second per member on CPU, and **Inference Memory** is negligible (tree inference is lightweight, not capped).

Evaluation & Benchmarking

Task for Performance Evaluation. Predict the integer age of each test crab and minimise MAE. **Performance Metrics.** MAE is the sole competition metric; our tree-search champion reaches **rounded OOF MAE 1.33563**, improving on the linear baseline's 1.33812 by 0.00249 (−0.19% relative). Both real leaderboard checks confirm CV trustworthiness:

#	Source	Rounded OOF MAE	Public LB	Private LB
1	Linear champion (LGB/XGB/CAT 0.6/0.3/0.1, exp #1)	1.33812	1.34356	1.34075
2	Tree-search best (node #15, 8-way blend, exp #5)	1.33563	1.34315	1.33859

Submission #2 beats #1 on **both** boards (Public −0.00041, Private −0.00216).

Performance Benchmarking. The natural comparator is the **NVIDIA reproduce-agent**, which verbatim-reproduces a strong public kernel:

Agent	Approach	MAE	Note
Our agent	from-scratch GBDT pool + tree-search	1.3356	winner
NVIDIA	verbatim reproduces <code>pandeyg0811/mae-1-337-with-featureengineering-xgb</code>	1.3387	−0.0031 behind

Our agent wins by 0.0031 MAE on this episode — the from-scratch pool with a rounded-metric-aware tree search edges out the reproduced public XGBoost kernel. The decisive levers were (1) precise blend-weight search decided directly on the rounded metric, and (2) genuinely diverse pool members — a feature-pruned LGB (dropping the $r=0.993$ collinear `Weight`) and a boundary-pushed low-LR LGB — that a verbatim kernel copy does not explore.